OBJECTIVE TESTS – 32

1. A transformer transforms
   (a) frequency
   (b) voltage
   (c) current
   (d) voltage and current.
2. Which of the following is not a basic element of a transformer ?
   (a) core
   (b) primary winding
   (c) secondary winding
   (d) mutual flux.
3. In an ideal transformer,
   (a) windings have no resistance
   (b) core has no losses
   (c) core has infinite permeability
   (d) all of the above.
4. The main purpose of using core in a transformer is to
   (a) decrease iron losses
   (b) prevent eddy current loss
   (c) eliminate magnetic hysteresis
   (d) decrease reluctance of the common magnetic circuit.
5. Transformer cores are laminated in order to
   (a) simplify its construction
   (b) minimise eddy current loss
   (c) reduce cost
   (d) reduce hysteresis loss.
6. A transformer having 1000 primary turns is connected to a 250-V a.c. supply. For a secondary voltage of 400 V, the number of secondary turns should be
   (a) 1600 (b) 250
   (c) 400 (d) 1250
7. The primary and secondary induced e.m.fs. E_1 and E_2 in a two-winding transformer are **always**
   (a) equal in magnitude
   (b) antiphase with each other
   (c) in-phase with each other
   (d) determined by load on transformer secondary.
8. A step-up transformer increases
   (a) voltage (b) current
   (c) power (d) frequency.
9. The primary and secondary windings of an ordinary 2-winding transformer **always** have
   (a) different number of turns
   (b) same size of copper wire
   (c) a common magnetic circuit
   (d) separate magnetic circuits.
10. In a transformer, the leakage flux of each winding is proportional to the current in that winding because
    (a) Ohm's law applies to magnetic circuits
    (b) leakage paths do not saturate
    (c) the two windings are electrically isolated
    (d) mutual flux is confined to the core.
11. In a two-winding transformer, the e.m.f. per turn in secondary winding is **always**.......the induced e.m.f. power turn in primary.
    (a) equal to K times
    (b) equal to $1/K$ times
    (c) equal to
    (d) greater than.
12. In relation to a transformer, the ratio 20 : 1 indicates that
    (a) there are 20 turns on primary one turn on secondary
    (b) secondary voltage is 1/20th of primary voltage
    (c) primary current is 20 times greater than the secondary current.
    (d) for every 20 turns on primary, there is one turn on secondary.
13. In performing the short circuit test of a transformer
    (a) high voltage side is usually short circuited
    (b) low voltage side is usually short circuited
    (c) any side is short circuited with preference
    (d) none of the above.
    (Elect. Machines, A.M.I.E. Sec. B, 1993)
14. The equivalent resistance of the primary of a transformer having $K = 5$ and $R_1 = 0.1$ ohm when referred to secondary becomes.......ohm.
    (a) 0.5
    (b) 0.02
    (c) 0.004
    (d) 2.5
15. A transformer has negative voltage regulation when its load power factor is
    (a) zero
    (b) unity
    (c) leading
    (d) lagging.

16. The primary reason why open-circuit test is performed on the low-voltage winding of the transformer is that it
(a) draws sufficiently large on-load current for convenient reading
(b) requires least voltage to perform the test
(c) needs minimum power input
(d) involves less core loss.

17. No-load test on a transformer is carried out to determine
(a) copper loss
(b) magnetising current
(c) magnetising current and no-load loss
(d) efficiency of the transformer.

18. The main purpose of performing open-circuit test on a transformer is to measure its
(a) Cu loss
(b) core loss
(c) total loss
(d) insulation resistance.

19. During short-circuit test, the iron loss of a transformer is negligible because
(a) the entire input is just sufficient to meet Cu losses only
(b) flux produced is a small fraction of the normal flux
(c) iron core becomes fully saturated
(d) supply frequency is held constant.

20. The iron loss of a transformer at 400 Hz is 10 W. Assuming that eddy current and hysteresis losses vary as the square of flux density, the iron loss of the transformer at rated voltage but at 50 Hz would be....... watt.
(a) 80 (b) 640
(c) 1.25 (d) 100

21. In operating a 400 Hz transformer at 50 Hz
(a) only voltage is reduced in the same proportion as the frequency
(b) only kVA rating is reduced in the same proportion as the frequency
(c) both voltage and kVA rating are reduced in the same proportion as the frequency
(d) none of the above.

22. The voltage applied to the h.v. side of a transformer during short-circuit test is 2% of its rated voltage. The core loss will be.......percent of the rated core loss.
(a) 4 (b) 0.4
(c) 0.25 (d) 0.04

23. Transformers are rated in kVA instead of kW because
(a) load power factor is often not known
(b) kVA is fixed whereas kW depends on load p.f.
(c) total transformer loss depends on volt-ampere
(d) it has become customary.

24. When a 400-Hz transformer is operated at 50 Hz its kVA rating is
(a) raduced to 1/8
(b) increased 8 times
(c) unaffected
(d) increased 64 times.

25. At relatively light loads, transformer efficiency is low because
(a) secondary output is low
(b) transformer losses are high
(c) fixed loss is high in proportion to the output
(d) Cu loss is small.

26. A 200 kVA transformer has an iron loss of 1 kW and full-load Cu loss of 2kW. Its load kVA corresponding to maximum efficiency is .... kVA.
(a) 100 (b) 141.4
(c) 50 (d) 200

27. If Cu loss of a transformer at 7/8th full load is 4900 W, then its full-load Cu loss would be ....watt.
(a) 5600 (b) 6400
(c) 375 (d) 429

28. The ordinary efficiency of a given transformer is maximum when
(a) it runs at half full-load
(b) it runs at full-load
(c) its Cu loss equals iron loss
(d) it runs slightly overload.

29. The output current corrèsponding to maximum efficiency for a transformer having core loss of 100 W and equivalent resistance referred to secondary of 0.25 Ω is .... ampere.
(a) 20
(b) 25
(c) 5
(d) 400

30. The maximum efficiency of a 100-kVA transformer having iron loss of 900 kW and F.L. Cu loss of 1600 W occurs at .... kVA.
(a) 56.3 (b) 133.3
(c) 75 (d) 177.7

31. The all-day efficiency of a transformer depends primarily on
(*a*) its copper loss
(*b*) the amount of load
(*c*) the duration of load
(*d*) both (*b*) and (*c*).

32. The marked increase in kVA capacity produced by connecting a 2 winding transformer as an autotransfomer is due to
(*a*) increase in turn ratio
(*b*) increase in secondary voltage
(*c*) increase in transformer efficiency
(*d*) establishment of conductive link between primary and secondary.

33. The kVA rating of an ordinary 2-winding transformer is increased when connected as an autotransformer because
(*a*) transformation ratio is increased
(*b*) secondary voltage is increased
(*c*) energy is transferred both inductively and conductivity
(*d*) secondary current is increased.

34. The saving in Cu achieved by converting a 2-winding transformer into an autotransformer is determined by
(*a*) voltage transformation ratio
(*b*) load on the secondary
(*c*) magnetic quality of core material
(*d*) size of the transformer core.

35. An autotransformer having a transformation ratio of 0.8 supplies a load of 3 kW. The power transferred conductively from primary to secondary is.......kW.
(*a*) 0.6 (*b*) 2.4
(*c*) 1.5 (*d*) 0.27

36. The essential condition for parallel opearation of two 1-ϕ transformers is that they should have the same
(*a*) polarity
(*b*) kVA rating
(*c*) voltage ratio
(*d*) percentage impedance.

37. If the impedance triangles of two transformers operating in parallel are not identical in shape and size, the two transformers will
(*a*) share the load unequally
(*b*) get heated unequally
(*c*) have a circulatory secondary current even when unloaded
(*d*) run with different power factors.

38. Two transformers *A* and *B* having equal outputs and voltage ratios but unequal percentage impedances of 4 and 2 are operating in parallel. Transformer *A* will be running over-load by .... percent.
(*a*) 50 (*b*) 66
(*c*) 33 (*d*) 25

ANSWERS

1. *d* **2.** *d* **3.** *d* **4.** *d* **5.** *b* **6.** *a* **7.** *c* **8.** *a* **9.** *c* **10.** *b* **11.** *c* **12.** *d* **13.** *b* **14.** *d* **15.** *c* **16.** *a* **17.** *c* **18.** *b* **19.** *b* **20.** *b* **21.** *b* **22.** *d* **23.** *c* **24.** *a* **25.** *c* **26.** *b* **27.** *b* **28.** *c* **29.** *a* **30.** *c* **31.** *d* **32.** *d* **33.** *c* **34.** *a* **35.** *b* **36.** *a* **37.** *d* **38.** *c*